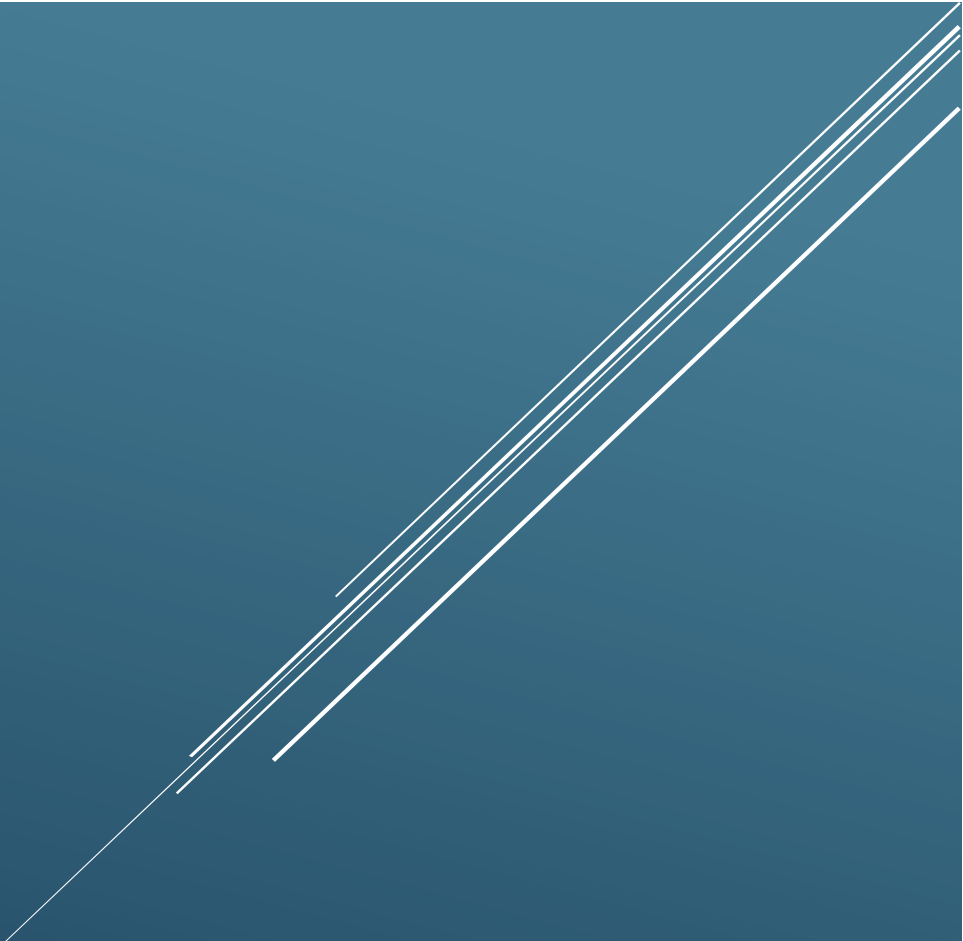




STELLARIS TECH SIMULATION ENVIRONMENT

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Executive Summary

This document provides an overview of the Stellaris Tech Simulation Environment, explaining its purpose, network architecture, virtual infrastructure, and the technologies that comprise the environment. Stellaris Tech simulates a small technology company, providing an environment for Security Operations Center (SOC) analysis, penetration testing, threat detection and hunting, vulnerability assessment, and network monitoring. It replicates common enterprise infrastructure components such as domain-joined user workstations, centralized authentication, security monitoring platforms, and segmented network architecture to reflect real-world operational conditions.

Simulation Scenario

Stellaris Tech is a fictional software development company responsible for developing and maintaining Stella Chronicles, a next-generation Virtual Reality Massive Multiplayer Online (VRMMO) platform. The organization operates a small enterprise network that supports financial operations, marketing, and IT administration. Because the company manages employee accounts, proprietary software, financial information, and other sensitive business data, maintaining the confidentiality, integrity, and availability of its systems is essential to daily operations.

Environment Overview

The simulation environment represents the enterprise infrastructure used by Stellaris Tech to support its business operations and cybersecurity program. The environment includes centralized identity management through Active Directory, departmental workstations, network segmentation, security monitoring platforms, vulnerability assessment tools, and intentionally vulnerable systems for offensive security testing. Together, these components create an environment for validating security controls, monitoring network activity, analyzing security events, and developing practical offensive and defensive cybersecurity skills.

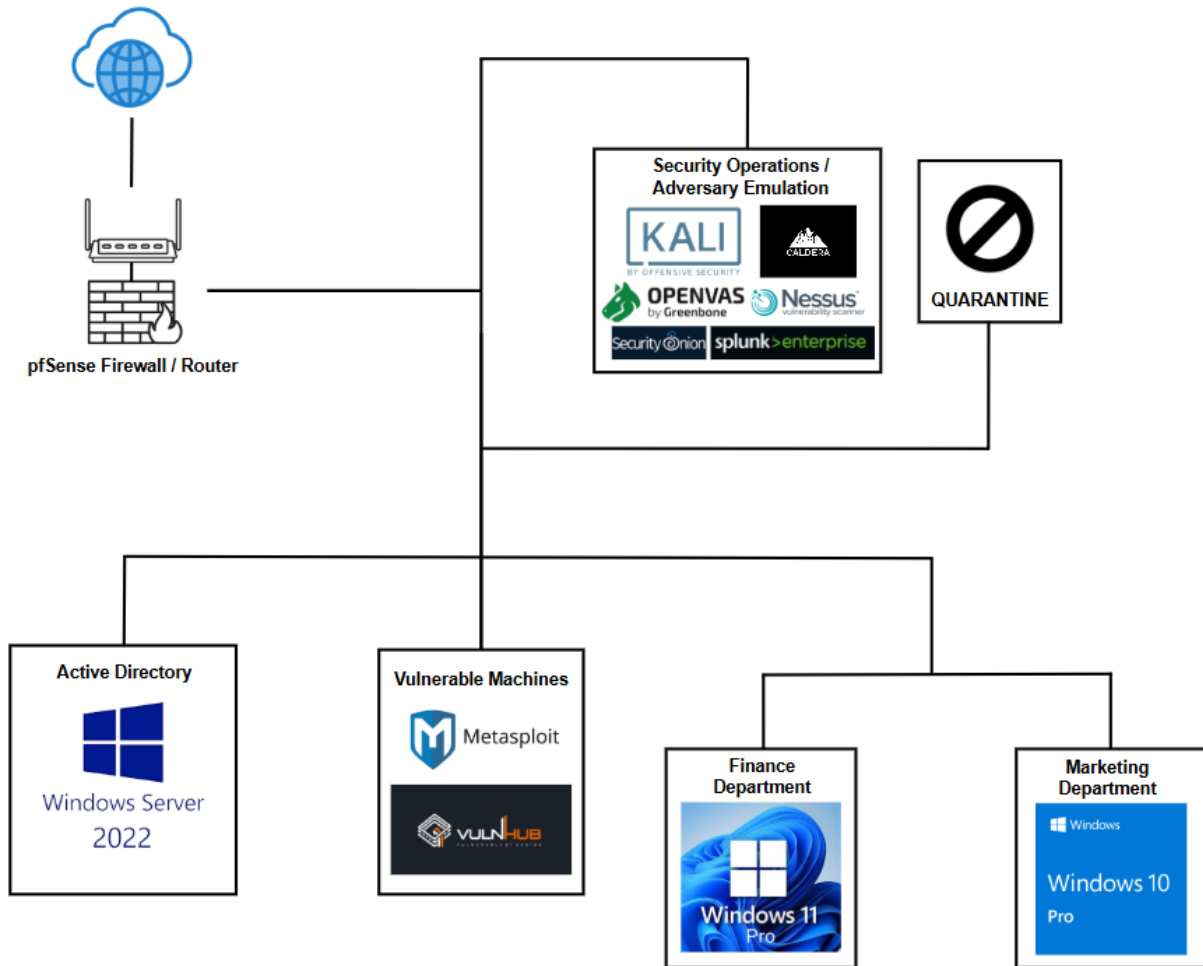
Simulation Environment Objectives

- Design and deploy a realistic small enterprise network that simulates a production business environment.
- Implement a segmented network architecture using pfSense to separate business systems, security infrastructure, and vulnerable machines.
- Build a centralized Active Directory domain to provide authentication, user management, and Group Policy administration.
- Deploy domain-joined Windows client workstations to represent organizational users and business operations.
- Integrate enterprise security platforms to support centralized logging, network monitoring, and adversary emulation.
- Deploy a system that aids in penetration testing.
- Establish a secure and scalable foundation that supports future simulation environment projects.

Host Machine Specifications

- CPU: Ryzen 9 5900x – 12 cores 24 threads
- GPU: MSI 3060TI 8GB DDR6
- RAM: Corsair Vengeance 32GB (8GB x 4) DDR4 at 3200 MHz
- Motherboard: Gigabyte A520 Aorus Elite
- PSU: EVGA 650 Watt Bronze
- Storage Drives: 1TB M.2 SSD (Main Drive) and 1TB SATA SSD (Dedicated for Simulation Environment)
- Virtualization Platform: VMware Workstation Pro

Virtual Infrastructure



Virtual Infrastructure Overview

The simulation environment infrastructure is designed to model the core systems commonly found in a small enterprise. It provides the foundation for deploying, managing, and securing business services while supporting cybersecurity operations in an isolated virtual environment.

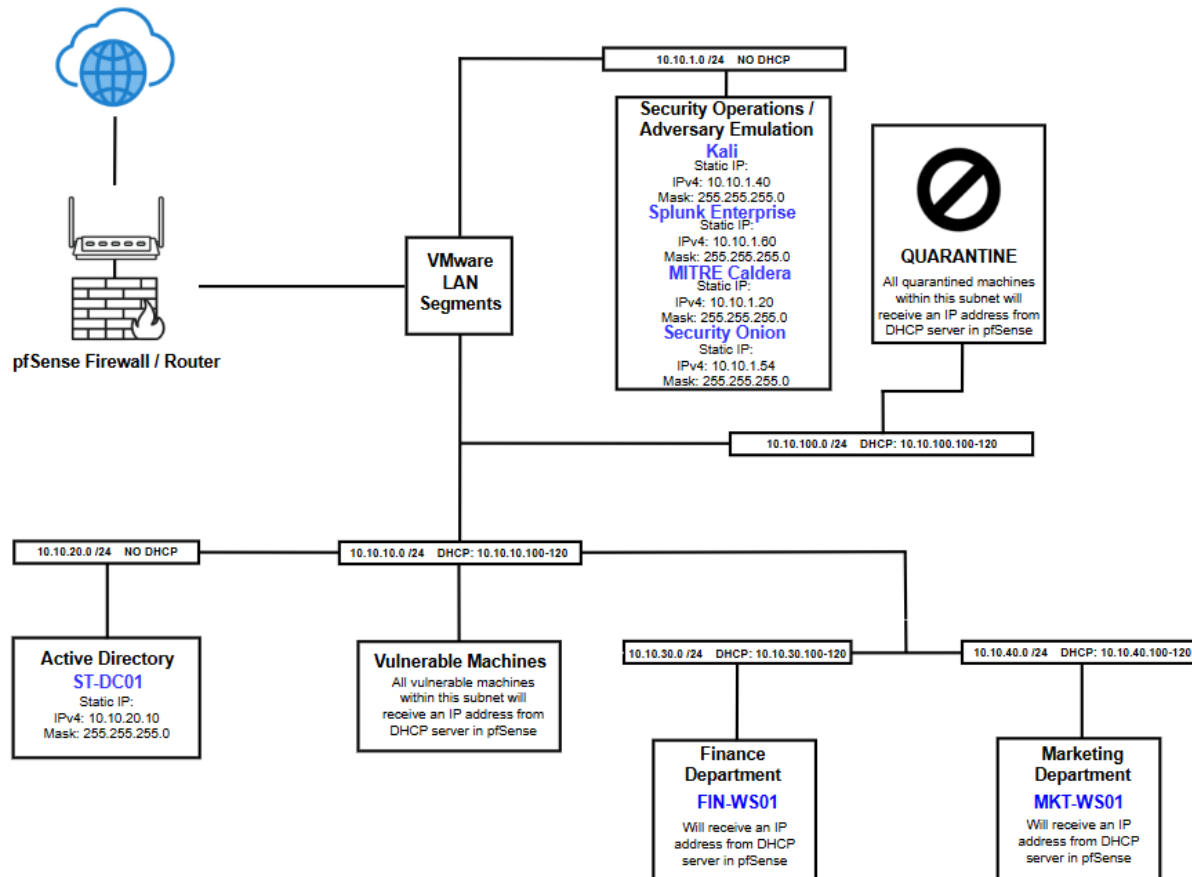
The infrastructure consists of virtual machines running VMware Workstation Pro, including a pfSense firewall, a Windows Server 2022 domain controller, Windows client workstations, security monitoring platforms, vulnerability assessment tools, adversary emulation systems, and intentionally vulnerable machines.

Virtual Infrastructure Components, Resource Allocation, and Purpose

Component / Resource Allocation	Purpose
VMware Workstation Pro	Hosts and manages the virtual simulation environment.
pfSense Resource Allocation: • OS: Ubuntu Server • CPU: 2 Processors • Memory: 2 GB • Storage: 30 GB	Provides routing, firewall, DHCP services, DNS services, and network segmentation.
Active Directory Domain Controller Resource Allocation: • OS: Windows Server 2022 • CPU: 2 Processors • Memory: 2 GB • Storage: 60 GB	Hosts Active Directory and acts as a centralized management system for user Identity and Access, grants file and application permissions, and enforces company-wide security policies.
Finance Workstation Resource Allocation: • OS: Windows 11 Pro • CPU: 2 Processors • Memory: 4 GB • Storage: 64 GB	Simulates a Finance Department workstation.
Marketing Workstation Resource Allocation: • OS: Windows 10 Pro • CPU: 2 Processors • Memory: 2 GB • Storage: 60 GB	Simulates a Marketing Department workstation.

Kali Linux Resource Allocation: • OS: Kali Linux • CPU: 4-6 GB (Varies on Computational Load) • Memory: 4-6 GB (Varies on Computational Load) • Storage: 120 GB	Used for vulnerability assessments and security testing.
Splunk Resource Allocation: • OS: Ubuntu Server • CPU: 4 Processors • Memory: 4 GB • Storage: 100 GB	Provides centralized log collection.
Security Onion Resource Allocation: • OS: Ubuntu Server • CPU: 4 Processors • Memory: 8 GB • Storage: 100 GB	Provides network monitoring and threat detection.
OpenVAS and Nessus Essentials Resource Allocation: • Shares resources from Kali Machine	Main tools for vulnerability assessments in the simulation environment. Two tools are provided for validating findings if either was used.
MITRE Caldera Resource Allocation: • OS: Ubuntu Server • CPU: 2 Processors • Memory: 4 GB • Storage: 100 GB	Allows adversary emulation against workstations within the simulation environment.
Metasploitable / VulnHub Resource Allocation: • OS: Varies • CPU: Varies • RAM: Varies • Storage: Varies	Serves as intentionally vulnerable systems for penetration testing.

Network Topology



Network Topology Overview

The simulation environments network is designed to reflect a segmented enterprise environment while providing a secure, flexible platform for cybersecurity operations. The topology utilizes pfSense as the central firewall and router, enabling communication between isolated network segments while enforcing network boundaries and access control.

The environment is divided into dedicated subnets based on system function. Separate network segments are used for Active Directory infrastructure, security operations and adversary emulation, vulnerable systems, business workstations, and quarantined hosts. This logical separation improves organization, limits unnecessary communication between systems, and enables security controls to be applied based on each subnet's role.

pfSense also provides DHCP, DNS, and network connectivity between the isolated network segments. VMware LAN Segments provide Layer 2 network isolation for the individual subnets. Static IP addressing is assigned to core infrastructure and security appliances to ensure consistent management, whereas business workstations and testing systems receive dynamic IP addresses through DHCP.

Network Design Considerations

The network architecture was designed within the capabilities of VMware Workstation Pro. Logical network segmentation is achieved using VMware LAN Segments, with each LAN Segment associated with a dedicated IP subnet. pfSense provides Layer 3 routing between the network segments and applies firewall policies to control communication between network zones. This approach provides functional network isolation while remaining portable, reproducible, and suitable for a desktop-based cybersecurity lab environment.

The simulation environment uses a structured IP addressing scheme based on the **10.10.x.0/24** private address space. Each subnet is assigned a unique third octet corresponding to its functional role within the environment, including Active Directory, Finance, Marketing, Security Operations, Quarantine, and Vulnerable Systems. This addressing convention was intentionally selected to provide a consistent, easy-to-remember format that simplifies network administration, troubleshooting, and documentation while reducing the likelihood of configuration errors.

Network Segments

Segment	Subnet	Gateway	Addressing Assignment	DHCP Scope	Primary Systems
Security Operations / Adversary Emulation	10.10.1.0 /24	10.10.1.254 /24	Static	N/A	• Kali Linux • Security Onion • Splunk Enterprise • OpenVAS • Nessus Essentials • MITRE Caldera
Vulnerable Machines	10.10.10.0 /24	10.10.10.254 /24	DHCP	10.10.10.50-100 /24	• VulnHub Machines • Metasploitable 2
Active Directory	10.10.20.0 /24	10.10.20.254 /24	Static	N/A	• Windows Server 2022
Finance Department	10.10.30.0 /24	10.10.30.254 /24	DHCP	10.10.30.100-120 /24	• Windows 11 Pro
Marketing Department	10.10.40.0 /24	10.10.40.254 /24	DHCP	10.10.40.100-120 /24	• Windows 10 Pro
Quarantine	10.10.100.0 /24	10.10.100.254 /24	DHCP	10.10.100.100-120 /24	• Any Quarantined VMs

Firewall and Routing

pfSense

pfSense serves as the primary network perimeter security device within the Stellaris Tech simulation environment. It functions as the firewall, router, and network segmentation control point, managing all inbound and outbound traffic between internal environment networks and external connections. pfSense uses firewall rules to isolate different parts of the environment, such as user workstations, servers, and security monitoring systems. It also provides essential networking services, such as a DHCP server.

The screenshot displays the pfSense Community Edition dashboard. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is divided into two columns. The left column, titled 'Available Widgets', contains a 'System Information' widget. The right column contains an 'Interfaces' widget and an 'Interface Statistics' widget.

System Information

Name	StellarisTech.home.arpa
User	admin@10.10.1.51 (Local Database)
System	VMware Virtual Machine Netgate Device ID: [REDACTED]
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Tue Feb 17 2026 Boot Method: BIOS
Version	2.8.1-RELEASE (amd64) built on Fri Oct 24 8:53:00 PDT 2025 FreeBSD 15.0-CURRENT The system is on the latest version. Version information updated at Sat Jul 4 16:21:16 PDT 2026
CPU Type	AMD Ryzen 9 5900X 12-Core Processor 2 CPUs : 2 package(s) x 1 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 14 Minutes 51 Seconds
Current date/time	Sat Jul 4 16:35:09 PDT 2026
DNS server(s)	• 127.0.0.1 • ::1

Interfaces

Interface	Speed	Link	IP Address
WAN	1000baseT <full-duplex>	↑	
SECURITYOPERATIONS	1000baseT <full-duplex>	↑	10.10.1.254
VULNERABLEMACHINES	1000baseT <full-duplex>	↑	10.10.10.254
ACTIVEDIRECTORY	1000baseT <full-duplex>	↑	10.10.20.254
SPAN	1000baseT <full-duplex>	↑	n/a
FINANCEDEPARTMENT	1000baseT <full-duplex>	↑	10.10.30.254
MARKETINGDEPARTMENT	1000baseT <full-duplex>	↑	10.10.40.254
QUARANTINE	1000baseT <full-duplex>	↑	10.10.100.254

Interface Statistics

	WAN	SECURITYOPERATIONS	VULNERABLEMACHINES	ACTIVEDIRECTORY
Packets In	10422	6426	0	0
Packets Out	8382	8820	5	5
Bytes In	10.35 MiB	473 KiB	0 B	0 B
Bytes Out	522 KiB	10.64 MiB	384 B	384 B
Errors In	0	0	0	0
Errors Out	0	0	0	0

Port Aliases

Port aliases were created in pfSense to simplify firewall rule management by grouping commonly used service ports into logical categories. Rather than specifying individual ports within each firewall rule, aliases allow multiple related ports to be referenced using a single name. This approach improves readability, reduces configuration complexity, and simplifies future maintenance by allowing port groups to be updated in a single location without modifying every associated firewall rule.

Firewall / Aliases / Ports

IP Ports URLs All

Firewall Aliases Ports				
Name	Type	Values	Description	Actions
DomainServices	Port(s)	53, 88, 123, 135, 389, 445, 464, 636, 3268, 3269...		  
MITRECaldera	Port(s)	8888, 7010, 80, 443, 445, 137, 138, 139		  
SecOnPorts	Port(s)	22, 80, 443, 4505, 4506, 5044, 8022, 514, 9001, 9003		  
SplunkPorts	Port(s)	9997, 8000, 8089		  
WebServices	Port(s)	80, 443, 8080, 53		  

Alias	Purpose
DomainServices	Groups the ports required for Active Directory services, including DNS, Kerberos, LDAP, SMB, and other authentication and directory service protocols.
MITRECaldera	Contains the ports used by the MITRE Caldera adversary emulation platform for agent communication and management.
SecOnPorts	Includes the ports required for Security Onion management, log collection, dashboards, and monitoring services.
SplunkPorts	Includes the ports required for Splunk forwarder for logs and management.
WebServices	Contains standard web service ports used for HTTP, HTTPS, DNS over web interfaces, and other web-based management applications.

The SPAN Port

To enable network-wide visibility, a dedicated Switched Port Analyzer (SPAN) interface was configured for Security Onion. Bridge interfaces were created for each network segment, including Security Operations, Active Directory, Finance, Marketing, Vulnerable Machines, and Quarantine, and connected to the SPAN interface. This configuration mirrors traffic from each subnet to Security Onion's monitoring interface without interrupting or altering normal network communications. As a result, Security Onion can inspect traffic across the entire environment.

Interfaces / Bridges

Interface Assignments

Interface Groups

Wireless

VLANs

QinQs

PPPs

GREs

GIFs

Bridges

LAGGs

Bridge Interfaces			
Interface	Members	Description	Actions
BRIDGE0	SECURITYOPERATIONS	Bridges SECOPS to SPAN	 
BRIDGE1	VULNERABLEMACHINES	Bridges VULNMACH to SPAN	 
BRIDGE2	ACTIVEDIRECTORY	Bridges AD to SPAN	 
BRIDGE3	FINANCEDEPARTMENT	Bridges FINDEP to SPAN	 
BRIDGE4	MARKETINGDEPARTMENT	Bridges MKTDEP to SPAN	 
BRIDGE5	QUARANTINE	Bridges QUARANTINE to SPAN	 

Firewall Rule Configurations

Firewall Design Philosophy

pfSense follows a default deny security model, meaning that any traffic not explicitly permitted by a firewall rule is automatically blocked. This approach enforces the principle of least privilege by allowing only authorized network communications while denying all other traffic. Throughout this environment, explicit pass or block rules were created only for services required by each subnet, allowing the implicit default deny policy to block all remaining traffic.

Firewall rules are also evaluated from top to bottom on each interface. Once a packet matches a rule, no additional rules are processed. For this reason, firewall rules are organized by function, with security monitoring and infrastructure services evaluated before broader Internet access rules. This ensures that specific traffic is handled appropriately before more general rules are considered.

Port aliases were also used to simplify firewall rule management by grouping related services under descriptive names. Using aliases improves readability, reduces configuration complexity, and allows firewall rules to be updated more efficiently without modifying individual entries.

WAN Interface Firewall Rules

Firewall rules are configured to:

- By default, pfSense has blocked outside private RFC 1918 and bogon IP networks from entering the network through the WAN interface.
- Block any outside source from accessing the firewall's GUI management interface.

Floating

WAN

SECURITYOPERATIONS

VULNERABLEMACHINES

ACTIVEDIRECTORY

SPAN

FINANCEDEPARTMENT

MARKETINGDEPARTMENT

QUARANTINE

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/1 KIB	*	RFC 1918 networks	*	*	*	*	*		Block private networks	
☒	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	
☐	0/280 B	IPv4 *	*	*	This Firewall (self)	*	*	none		Blocks Firewall GUI Access Outside The Network	

Security Operations Interface Firewall Rules

Firewall rules are configured to:

- Allow needed ports from Splunk and Security Onion from the Security Operations subnets to the network.
- Allow the Security Operations subnet to communicate with all other internal subnets.
- Allow needed ports from MITRE Caldera to the network.
- Allow ICMP traffic from the Security Operations subnet for network diagnostics.
- Allow web services for the Security Operations subnet.

Floating	WAN	SECURITYOPERATIONS	VULNERABLEMACHINES	ACTIVEDIRECTORY	SPAN	FINANCEDEPARTMENT
MARKETINGDEPARTMENT	QUARANTINE					

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1/1.07 MiB	*	*	*	SECURITYOPERATIONS Address	443 80	*	*		Anti-Lockout Rule	
Security Monitoring											
☐	☒ 0/0 B	IPv4 TCP	SECURITYOPERATIONS subnets	*	*	SplunkPorts	*	none		Connects Needed Splunk Ports form SecOps Subnet to Network	
☐	☒ 13/51.92 MiB	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	*	SecOnPorts	*	none		Connects Needed Security Onion Ports From SecOps To Network	
Infrastructure Management											
☐	☒ 0/0 B	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	ACTIVEDIRECTORY subnets	*	*	none		Allows Connection From SecOps to AD Subnets	
☐	☒ 0/0 B	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	FINANCEDEPARTMENT subnets	*	*	none		Allows Connection From SecOps to FIN Subnets	
☐	☒ 0/0 B	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	MARKETINGDEPARTMENT subnets	*	*	none		Allows Connection From SecOps to MKT Subnets	
☐	☒ 0/0 B	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	VULNERABLEMACHINES subnets	*	*	none		Allows Connection From SecOps to VulnMach Subnets	
☐	☒ 0/0 B	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	QUARANTINE subnets	*	*	none		Allows Connection From SecOps to Quarantine Subnets	
Adversary Emulation											
☐	☒ 0/0 B	IPv4 TCP	SECURITYOPERATIONS subnets	*	*	MITRECaldera	*	none		Connects Needed MITRE Caldera Ports From SecOps To Network	
Network Diagnostics											
☐	☒ 0/0 B	IPv4 ICMP any	SECURITYOPERATIONS subnets	*	*	*	*	none			
Internet Access											
☐	☒ 1/211 KiB	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	*	WebServices	*	none		Internet for Security Operations Subnet	

Active Directory Interface Firewall Rules

Firewall rules are configured to:

- Block the Active Directory subnet from accessing the firewall interface.
- Block ICMP traffic from the Active Directory subnet to the Security Operations, Vulnerable Machines, and Quarantine subnets.
- Allow the Active Directory subnet to access the needed Splunk and Security Onion ports from the Security Operations subnet.
- Allow the Active Directory subnet to give domain services to the Financial and Marketing Department subnets.
- Allow ICMP traffic from the Active Directory subnet for network diagnostics.
- Allow web services for the Active Directory subnet.

Floating

WAN

SECURITYOPERATIONS

VULNERABLEMACHINES

ACTIVEDIRECTORY

SPAN

FINANCEDEPARTMENT

MARKETINGDEPARTMENT

QUARANTINE

Rules (Drag to Change Order)

States

Protocol

Source

Port

Destination

Port

Gateway

Queue

Schedule

Description

Actions

BLOCK

✗

0/0 B

IPv4 TCP

ACTIVEDIRECTORY subnets

*

This Firewall (self)

*

*

none

Block AD From Accessing Firewall Web GUI

✗

0/0 B

IPv4 ICMP any

ACTIVEDIRECTORY subnets

*

SECURITYOPERATIONS subnets

*

*

none

Block ICMP FROM FIN to SecOps Subnet

✗

0/0 B

IPv4 ICMP any

ACTIVEDIRECTORY subnets

*

VULNERABLEMACHINES subnets

*

*

none

Block ICMP FROM AD to VulnMach Subnet

✗

0/0 B

IPv4 ICMP any

ACTIVEDIRECTORY subnets

*

QUARANTINE subnets

*

*

none

Block ICMP FROM AD to Quarantine Subnet

Security Monitoring

✓

0/53.46 MiB

IPv4 TCP

ACTIVEDIRECTORY subnets

*

SECURITYOPERATIONS subnets

SplunkPorts

*

none

Connects Needed Splunk Ports form AD Subnet to SecOps Subnet

✓

0/0 B

IPv4 TCP/UDP

ACTIVEDIRECTORY subnets

*

SECURITYOPERATIONS subnets

SecOnPorts

*

none

Connects Needed Security Onion Ports To AD Subnet

Domain Controller

✓

0/0 B

IPv4 TCP/UDP

ACTIVEDIRECTORY subnets

*

FINANCEDEPARTMENT subnets

DomainServices

*

none

Gives Domain Services From AD to Finance Department

✓

0/0 B

IPv4 TCP/UDP

ACTIVEDIRECTORY subnets

*

MARKETINGDEPARTMENT subnets

DomainServices

*

none

Gives Domain Services From AD to Marketing Department

Adversary Emulation

✓

0/0 B

IPv4 TCP

ACTIVEDIRECTORY subnets

*

SECURITYOPERATIONS subnets

MITRECaldera

*

none

Connects Needed MITRE Caldera Ports TO AD Subnet

Network Diagnostics

✓

0/0 B

IPv4 ICMP any

ACTIVEDIRECTORY subnets

*

*

*

*

none

Allows ICMP From AD

Internet Access

✓

0/14.38 MiB

IPv4 TCP/UDP

ACTIVEDIRECTORY subnets

*

*

WebServices

*

none

Internet for Active Directory Subnet

Finance Department Interface Firewall Rules

Firewall rules are configured to:

- Block the Finance Department subnet from accessing the firewall interface.
- Block ICMP traffic from the Finance Department subnet to the Security Operations, Vulnerable Machines, and Quarantine subnets.
- Allow the Finance Department subnet to receive the needed Splunk and Security Onion ports from the Security Operations subnet.
- Allow the Finance Department subnet to receive domain services from the Active Directory subnet.
- Allow ICMP traffic from the Finance Department subnet for network diagnostics.
- Allow web services for the Finance Department subnet.

Floating

WAN

SECURITYOPERATIONS

VULNERABLEMACHINES

ACTIVEDIRECTORY

SPAN

FINANCEDEPARTMENT

MARKETINGDEPARTMENT

QUARANTINE

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
BLOCK											
☐		0/0 B	IPv4 TCP	FINANCEDEPARTMENT subnets	*	This Firewall (self)	*	*	none	Block FIN From Accessing Firewall Web GUI	
☐		0/0 B	IPv4 ICMP any	FINANCEDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	*	*	none	Block ICMP FROM FIN to SecOps Subnet	
☐		0/0 B	IPv4 ICMP any	FINANCEDEPARTMENT subnets	*	VULNERABLEMACHINES subnets	*	*	none	Block ICMP FROM FIN to VulnMach Subnet	
☐		0/0 B	IPv4 ICMP any	FINANCEDEPARTMENT subnets	*	QUARANTINE subnets	*	*	none	Block ICMP FROM FIN to Quarantine Subnet	
Security Monitoring											
☐		0/0 B	IPv4 TCP	FINANCEDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	SplunkPorts	*	none	Connects Needed Splunk Ports form FIN Subnet to SecOps Subnet	
☐		0/0 B	IPv4 TCP/UDP	FINANCEDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	SecOnPorts	*	none	Connects Needed Security Onion Ports To FIN Subnet	
Domain Controller											
☐		0/0 B	IPv4 TCP/UDP	FINANCEDEPARTMENT subnets	*	ACTIVEDIRECTORY subnets	DomainServices	*	none	Domain Authentication	
Adversary Emulation											
☐		0/0 B	IPv4 TCP	FINANCEDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	MITRECaldera	*	none	Connects Needed MITRE Caldera Ports To FIN Subnet	
Network Diagnostics											
☐		0/0 B	IPv4 ICMP any	FINANCEDEPARTMENT subnets	*	*	*	*	none	Allows ICMP From Finance Department	
Internet Access											
☐		0/0 B	IPv4 TCP/UDP	FINANCEDEPARTMENT subnets	*	*	WebServices	*	none	Internet for Finance Department	

Marketing Department Interface Firewall Rules

Firewall rules are configured to:

- Block the Marketing Department subnet from accessing the firewall interface.
- Block ICMP traffic from the Marketing Department subnet to the Security Operations, Vulnerable Machines, and Quarantine subnets.
- Allow the Marketing Department subnet to receive the needed Splunk and Security Onion ports from the Security Operations subnet.
- Allow the Marketing Department subnet to receive domain services from the Active Directory subnet.
- Allow ICMP traffic from the Marketing Department subnet for network diagnostics.
- Allow web services for the Marketing Department subnet.

Floating	WAN	SECURITYOPERATIONS	VULNERABLEMACHINES	ACTIVEDIRECTORY	SPAN	FINANCEDEPARTMENT
MARKETINGDEPARTMENT	QUARANTINE					

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
BLOCK											
☐	✗	0/0 B	IPv4 TCP	MARKETINGDEPARTMENT subnets	*	This Firewall (self)	*	*	none	Block MKT From Accessing Firewall Web GUI	
☐	✗	0/0 B	IPv4 ICMP any	MARKETINGDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	*	*	none	Block ICMP FROM MKT to SecOps Subnet	
☐	✗	0/0 B	IPv4 ICMP any	MARKETINGDEPARTMENT subnets	*	VULNERABLEMACHINES subnets	*	*	none	Block ICMP FROM MKT to VulnMach Subnet	
☐	✗	0/0 B	IPv4 ICMP any	MARKETINGDEPARTMENT subnets	*	QUARANTINE subnets	*	*	none	Block ICMP FROM MKT to Quarantine Subnet	
Security Monitoring											
☐	✓	0/0 B	IPv4 TCP	MARKETINGDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	SplunkPorts	*	none	Connects Needed Splunk Ports form MKT Subnet to SecOps Subnet	
☐	✓	0/0 B	IPv4 TCP/UDP	MARKETINGDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	SecOnPorts	*	none	Connects Needed Security Onion Ports To MKT Subnet	
Infrastructure Management											
☐	✓	0/0 B	IPv4 TCP/UDP	MARKETINGDEPARTMENT subnets	*	ACTIVEDIRECTORY subnets	DomainServices	*	none	Domain Authentication	
Adversary Emulation											
☐	✓	0/0 B	IPv4 TCP	MARKETINGDEPARTMENT subnets	*	SECURITYOPERATIONS subnets	MITRECaldera	*	none	Connects Needed MITRE Caldera Ports To MKT Subnet	
Network Diagnostics											
☐	✓	0/0 B	IPv4 ICMP any	MARKETINGDEPARTMENT subnets	*	*	*	*	none	Allows ICMP From Marketing Department	
Internet Access											
☐	✓	0/0 B	IPv4 TCP/UDP	MARKETINGDEPARTMENT subnets	*	*	WebServices	*	none	Internet for Marketing Department	

Vulnerable Machines Interface Firewall Rules

Firewall rules are configured to:

- Block the Vulnerable Machines subnet from accessing the firewall interface.
- Allow the Vulnerable Machines subnet to receive the needed Splunk and Security Onion ports from the Security Operations subnet.

Floating	WAN	SECURITYOPERATIONS	VULNERABLEMACHINES	ACTIVEDIRECTORY	SPAN	FINANCEDEPARTMENT
MARKETINGDEPARTMENT	QUARANTINE					

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
BLOCK											
☐	✗	0/0 B	IPv4 TCP	VULNERABLEMACHINES subnets	*	This Firewall (self)	*	*	none	Block VulnMach From Accessing Firewall Web GUI	
Security Monitoring											
☐	✓	0/0 B	IPv4 TCP	VULNERABLEMACHINES subnets	*	SECURITYOPERATIONS subnets	SplunkPorts	*	none	Connects Needed Splunk Ports form VulnMach Subnet to SecOps Subnet	
☐	✓	0/0 B	IPv4 TCP/UDP	VULNERABLEMACHINES subnets	*	SECURITYOPERATIONS subnets	SecOnPorts	*	none	Connects Needed Security Onion Ports To VulnMach Subnet	

Quarantine Interface Firewall Rules

Firewall rules are configured to:

- Block the Quarantine subnet from accessing the firewall interface.
- Allow the Quarantine subnet to receive the needed Splunk and Security Onion ports from the Security Operations subnet.
- Allow web services for the Quarantine subnet to obtain required updates.

Floating	WAN	SECURITYOPERATIONS	VULNERABLEMACHINES	ACTIVEDIRECTORY	SPAN	FINANCEDEPARTMENT
MARKETINGDEPARTMENT	QUARANTINE					

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
BLOCK											
☐	✗	0/0 B	IPv4 TCP	QUARANTINE subnets	*	This Firewall (self)	*	*	none	Block Quarantine From Accessing Firewall Web GUI	
Security Monitoring											
☐	✓	0/0 B	IPv4 TCP	QUARANTINE subnets	*	SECURITYOPERATIONS subnets	SplunkPorts	*	none	Connects Needed Splunk Ports form Quarantine Subnet to SecOps Subnet	
☐	✓	0/0 B	IPv4 TCP/UDP	QUARANTINE subnets	*	SECURITYOPERATIONS subnets	SecOnPorts	*	none	Connects Needed Security Onion Ports To Quarantine Subnet	
Internet Access For Updates											
☐	✓	0/0 B	IPv4 TCP/UDP	SECURITYOPERATIONS subnets	*	*	WebServices	*	none	Internet for Quarantine Subnet For Updates Only	

SPAN Interface Firewall Rules

The SPAN interface requires no rules; it exists solely to mirror traffic from other interfaces to Security Onion.

Floating

WAN

SECURITYOPERATIONS

VULNERABLEMACHINES

ACTIVEDIRECTORY

SPAN

FINANCEDEPARTMENT

MARKETINGDEPARTMENT

QUARANTINE

Rules (Drag to Change Order)

States

Protocol

Source

Port

Destination

Port

Gateway

Queue

Schedule

Description

Actions

No rules are currently defined for this interface

All incoming connections on this interface will be blocked until pass rules are added. Click the button to add a new rule.

Active Directory

Windows Server 2022 (ST-DC01)

The Active Directory infrastructure serves as the centralized identity and access management solution for the Stellaris Tech simulation environment. Built on Windows Server 2022, it provides centralized authentication, authorization, and policy management for domain-joined systems, creating a realistic enterprise network for cybersecurity testing. The domain is organized into departmental Organizational Units (OUs) for Finance, Marketing, IT, Servers, and Workstations to simulate a structured business environment and simplify administrative management. Additionally, Sysmon is installed on the domain controller to enhance endpoint visibility by recording detailed system activity, including process creation, network connections, file operations, and registry modifications, supporting security monitoring and forensic analysis.

The screenshot displays the Windows Server Manager interface for a local server named ST-DC01. The left-hand navigation pane includes links to the Dashboard, Local Server, All Servers, AD DS, DHCP, DNS, and File and Storage Services. The main area is divided into two sections: PROPERTIES and EVENTS.

PROPERTIES For ST-DC01

Property	Value	Property	Value
Computer name	ST-DC01	Last installed updates	6/8/2026
Domain	stellaristech.local	Windows Update	Download
		Last checked for updates	Yesterday
Microsoft Defender Firewall	Public: On	Microsoft Defender Antivirus	Real-Time
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	Off
NIC Teaming	Disabled	Time zone	(UTC-08:00)
Ethernet0	10.10.20.10, IPv6 enabled	Product ID	00454-40
Azure Arc Management	Disabled		
Operating system version	Microsoft Windows Server 2022 Standard Evaluation	Processors	AMD Ryz
Hardware information	VMware, Inc. VMware20,1	Installed memory (RAM)	2 GB
		Total disk space	50.77 GB

EVENTS
All events | 24 total

Filter [] [] [] []

Server Name	ID	Severity	Source	Log	Date and Time
ST-DC01	10016	Warning	Microsoft-Windows-DistributedCOM	System	8/2/2026 7:54:31 PM
ST-DC01	10016	Warning	Microsoft-Windows-DistributedCOM	System	8/2/2026 7:54:18 PM
ST-DC01	12	Warning	Microsoft-Windows-Time-Service	System	8/2/2026 7:53:56 PM

Setting a Static IP and DNS Address for the Domain Controller

The Windows Server 2022 Domain Controller is configured with a static IP address to provide a consistent and reliable endpoint for Active Directory services. Because the server hosts the Active Directory Domain Services (AD DS) role and integrated DNS, its network address must remain consistent to ensure reliable communication between domain-joined systems. The DNS service maintains records that allow domain-joined clients to locate the Domain Controller and other Active Directory services. pfSense serves as the network gateway and firewall for the environment, while the Domain Controller handles DNS resolution for the Active Directory domain.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 10 . 20 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 10 . 10 . 20 . 254

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 10 . 10 . 20 . 254

Alternate DNS server: 8 . 8 . 8 . 8

☐ Validate settings upon exit

Advanced...

OK Cancel

Stellaris Tech Organizational Unit Structure

The Active Directory environment is organized into a hierarchical Organizational Unit (OU) structure that simplifies administration and reflects Stellaris Techs' organizational layout. User accounts, computer objects, security groups, and administrative accounts are separated into dedicated OUs, allowing resources to be managed independently while supporting future scalability and policy enforcement.

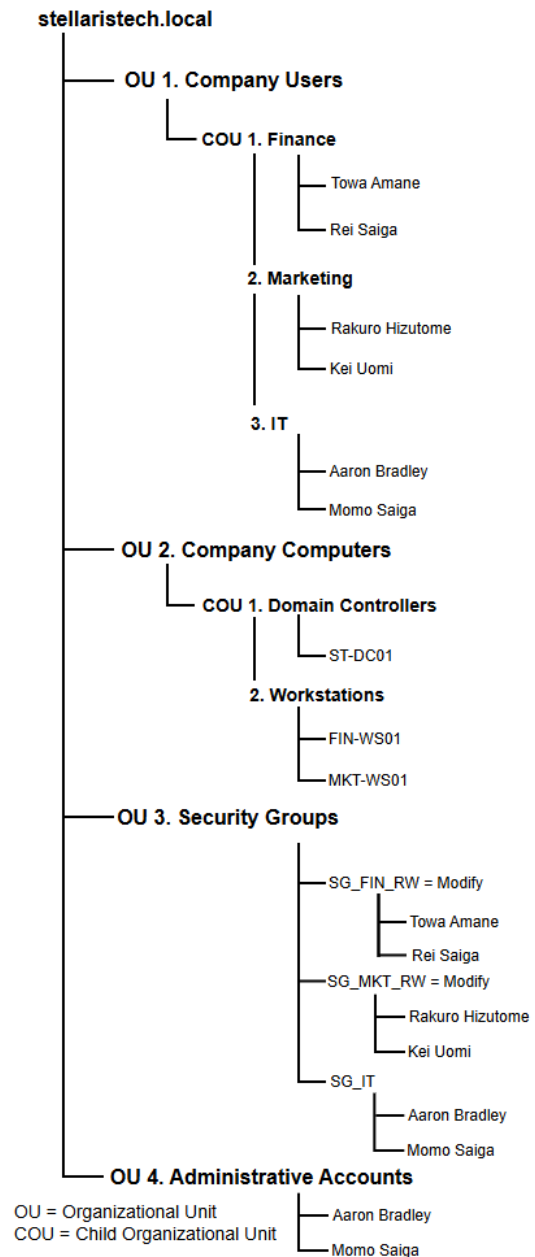
The **Company Users OU** contains child OUs for the Finance, Marketing, and IT departments, with user accounts organized according to their respective business functions.

The **Company Computers OU** is divided into Domain Controllers and Workstations, separating server infrastructure from client systems. The domain controller (ST-DC01) is isolated within its own child OU, while department workstations (FIN-WS01 and MKT-WS01) are grouped under the Workstations OU.

The **Security Groups OU** contains role-based security groups that manage access permissions for each department. Finance and Marketing users are assigned to department-specific read/write groups, while IT personnel are assigned to a dedicated IT security group. Managing permissions through security groups rather than individual user accounts simplifies administration and aligns with the principle of least privilege.

The **Administrative Accounts OU** stores dedicated privileged accounts separately from standard user accounts. This separation supports security best practices by distinguishing administrative activities from normal user operations and reducing the need for elevated privileges for routine tasks.

Stellaris Tech OU Structure

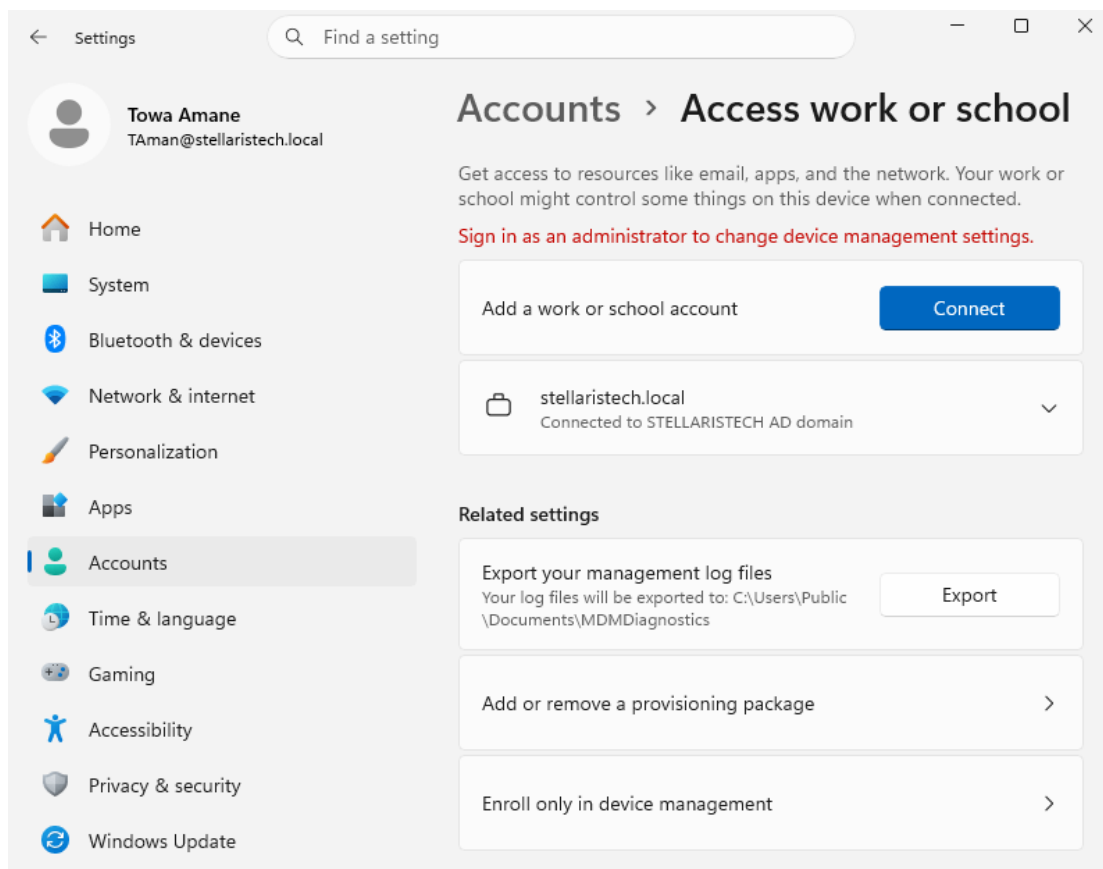


Finance Department

FIN-WS01

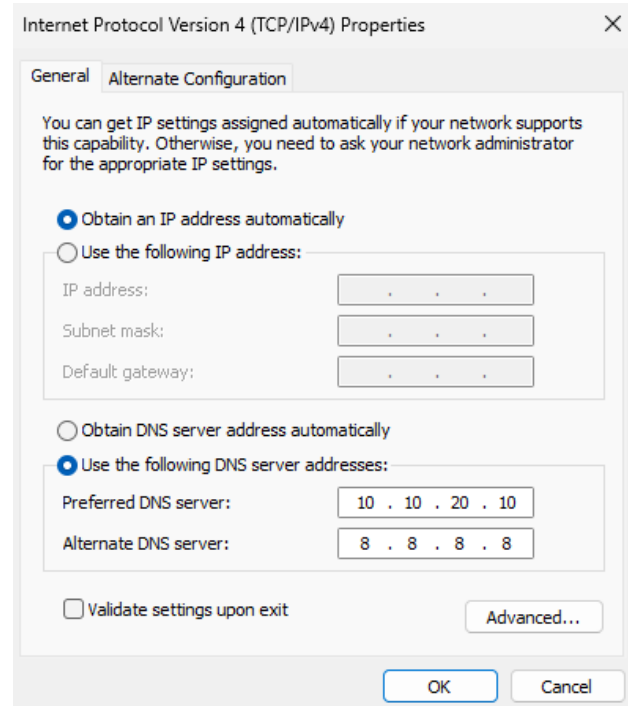
FIN-WS01 is a Windows 11 Pro workstation that simulates a finance department endpoint within the Stellaris Tech environment. As a domain-joined client, it participates in centralized authentication, Group Policy enforcement, and user account management through Active Directory. The workstation is designed to simulate a business-critical system that handles sensitive financial operations while generating realistic Windows event logs and network traffic for security monitoring. Sysmon is installed to provide enhanced endpoint telemetry by recording detailed system activity, including process creation, network connections, file operations, and registry modifications, supporting security monitoring, threat detection, and forensic analysis. The workstation also serves as a target for vulnerability assessments, security configuration validation, and defensive security exercises using the tools deployed throughout the environment.

Towa Amane, an employee within the Finance department, with the FIN-WS01 connected to the Domain Controller, is shown below:



Pointing FIN-WS01 to the Domain Controllers DNS

FIN-WS01 is configured to use the domain controller (ST-DC01) as its preferred DNS server because Active Directory relies on DNS to locate and access domain services. By pointing the workstation to the domain controller at 10.10.20.10, FIN-WS01 can resolve the internal stellaristech.local domain and locate services such as Kerberos and LDAP through Active Directory DNS records. This allows the workstation to authenticate users, apply Group Policy Objects (GPOs), and communicate with other domain resources. The workstation uses the domain controller as its only DNS server to ensure that internal domain-related queries are consistently resolved through the Active Directory DNS service. External DNS requests can be handled by configuring a DNS forwarder on the domain controller rather than assigning a public DNS server directly to the workstation.

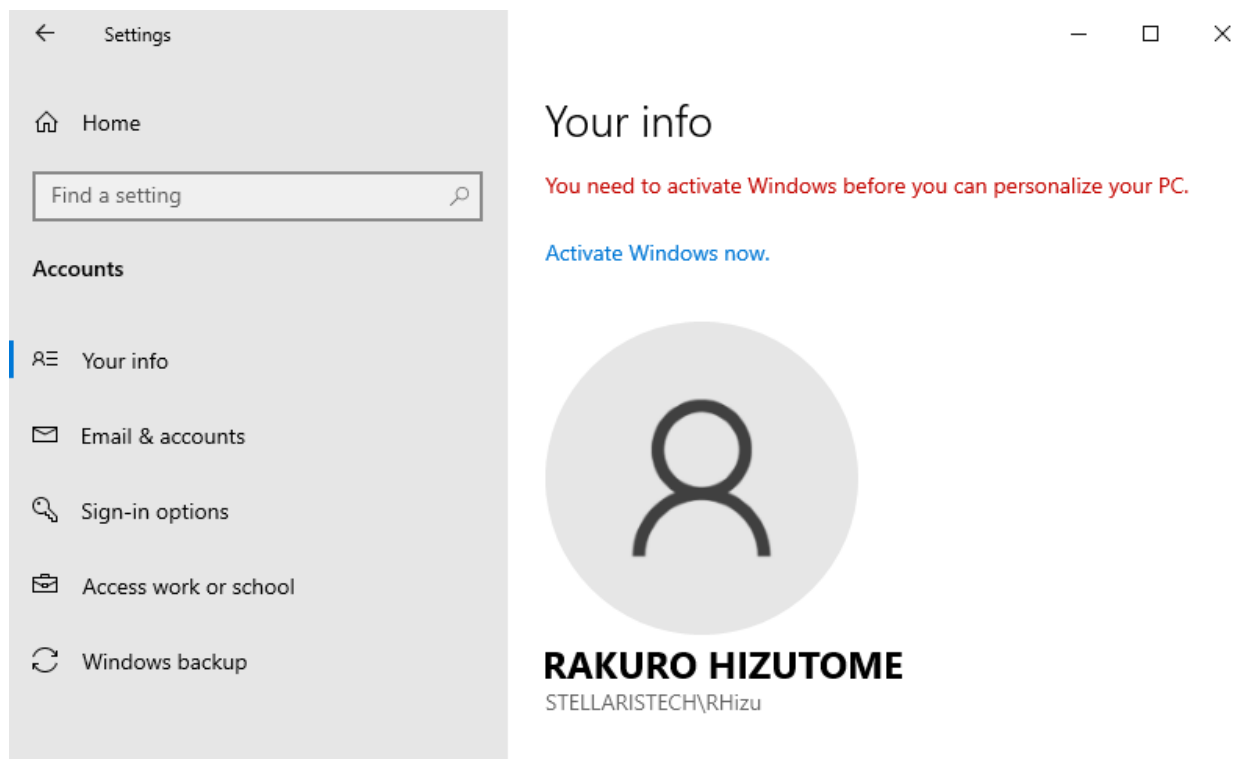


Marketing Department

MKT-WS01

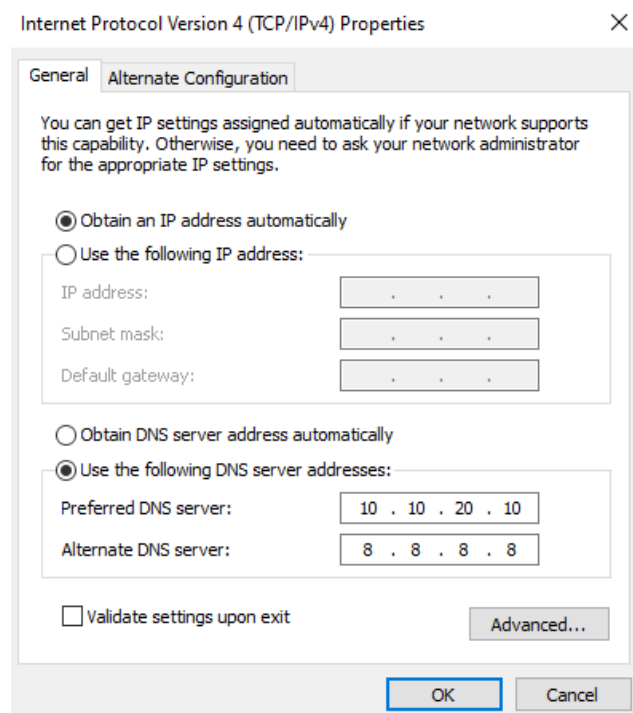
MKT-WS01 is a Windows 10 Pro workstation that simulates a marketing department endpoint within the Stellaris Tech environment. Windows 10 is used to represent organizations that continue to support legacy operating systems due to application compatibility, hardware limitations, or phased operating system upgrade strategies commonly found in enterprise environments. As a domain-joined client, the workstation participates in centralized authentication, Group Policy enforcement, and user account management through Active Directory. Sysmon is installed to provide enhanced endpoint telemetry by recording detailed system activity, including process creation, network connections, file operations, and registry modifications, supporting security monitoring, threat detection, and forensic analysis. The workstation also serves as a target for vulnerability assessments, security configuration validation, and defensive security exercises using the tools deployed throughout the environment.

Rakuro Hizutome, an employee within the Marketing Department, with MKT-WS01 connected to the Domain Controller, is shown below:



Pointing MKT-WS01 to the Domain Controllers DNS

MKT-WS01 is configured to use the domain controller (ST-DC01) as its preferred DNS server because Active Directory relies on DNS to locate and access domain services. By pointing the workstation to the domain controller at 10.10.20.10, MKT-WS01 can resolve the internal stellaristech.local domain and locate services such as Kerberos and LDAP through Active Directory DNS records. This allows the workstation to authenticate users, apply Group Policy Objects (GPOs), and communicate with other domain resources. The workstation uses the domain controller as its only DNS server to ensure that internal domain-related queries are consistently resolved through the Active Directory DNS service. External DNS requests can be handled by configuring a DNS forwarder on the domain controller rather than assigning a public DNS server directly to the workstation.



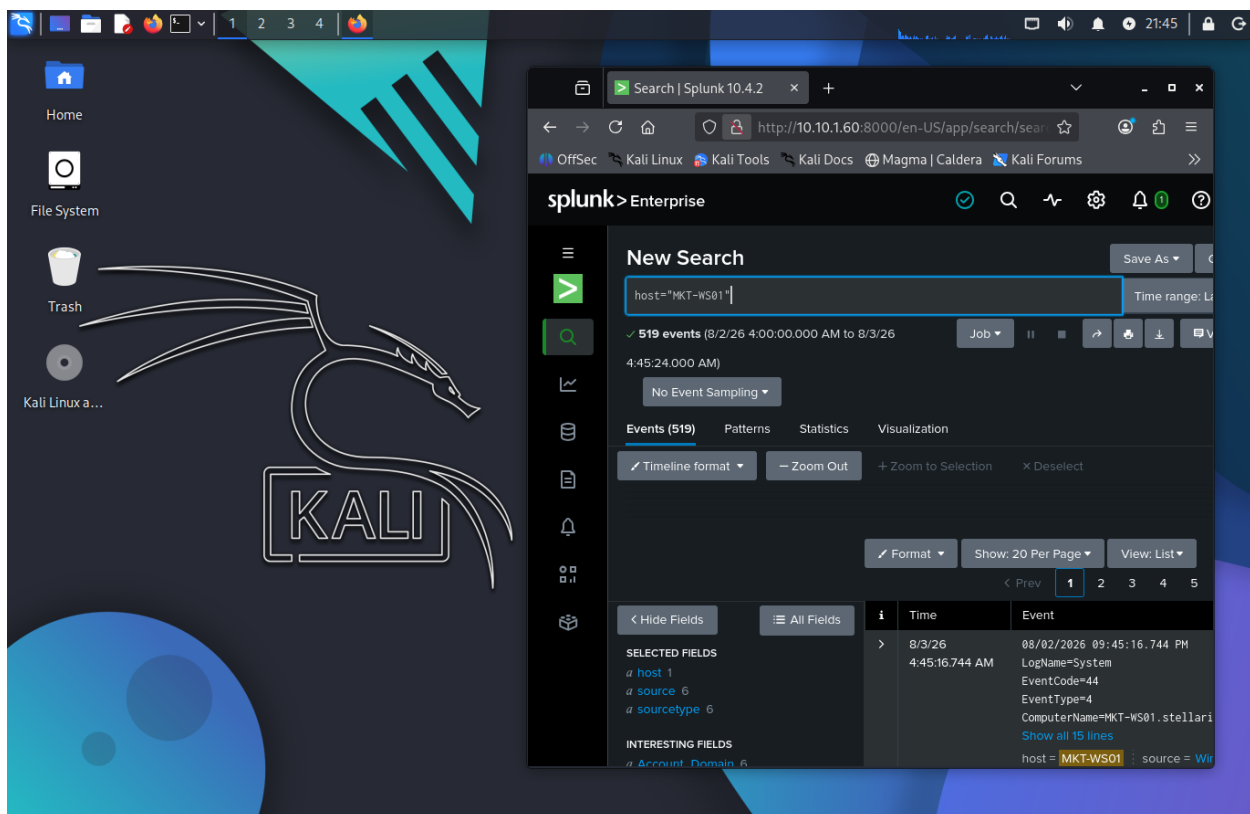
The screenshot shows the 'Internet Protocol Version 4 (TCP/IPv4) Properties' dialog box with the 'Alternate Configuration' tab selected. The dialog box contains the following elements:

- General** and **Alternate Configuration** tabs.
- Introductory text: "You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings."
- Radio button selection for IP address assignment:
 - ☒ Obtain an IP address automatically
 - ☐ Use the following IP address:
- Fields for manual IP configuration (when the second radio button is selected):
 - IP address: [. . .]
 - Subnet mask: [. . .]
 - Default gateway: [. . .]
- Radio button selection for DNS server address assignment:
 - ☐ Obtain DNS server address automatically
 - ☒ Use the following DNS server addresses:
- Fields for manual DNS configuration (when the second radio button is selected):
 - Preferred DNS server: [10 . 10 . 20 . 10]
 - Alternate DNS server: [8 . 8 . 8 . 8]
- ☐ Validate settings upon exit
- Advanced...** button
- OK** and **Cancel** buttons at the bottom.

Security Operations and Adversary Emulation

Kali Linux

Kali Linux serves as the primary administrative and security assessment workstation within the Stellaris Tech simulation environment. It provides a centralized platform for managing and accessing the environment's security infrastructure, including pfSense, OpenVAS, Nessus, Splunk, Security Onion, and MITRE Caldera. Kali Linux is primarily used to conduct vulnerability assessments, security testing, and reconnaissance against systems within the environment, while also providing administrative access to security platforms hosted on separate servers. MITRE Caldera is hosted independently and is accessed from Kali Linux when conducting adversary emulation activities. This separation allows Kali Linux to function as the primary security assessment workstation while dedicated servers perform specialized security monitoring and adversary emulation functions.



Splunk Enterprise

Splunk is a Security Information and Event Management (SIEM) platform used to collect, index, and analyze log data from systems, applications, and network devices. Within the Stellaris Tech simulation environment, Splunk serves as the primary centralized logging and security monitoring solution by aggregating logs from Active Directory, workstations, and pfSense. Splunk enables real-time monitoring, event correlation, custom dashboards, and alerting to support threat detection, incident investigation, and security operations. It also provides historical log retention and powerful search capabilities that assist with identifying suspicious activity and validating security events across the environment. In addition to collecting live telemetry from the simulation environment, Splunk supports the use of Splunk Boss of the SOC (BOTS) datasets, allowing realistic attack scenarios and Advanced Persistent Threat (APT) activity to be analyzed using Search Processing Language (SPL). This provides additional opportunities to develop threat hunting, detection engineering, and incident response skills using real-world security datasets.

AD server and workstations connected to Splunk Enterprise with indexed data shown below:

The image displays two screenshots of the Splunk Enterprise interface. The top screenshot shows a 'Data Summary' dashboard with a table of indexed data. The bottom screenshot shows a 'New Search' view with a search query and a list of events.

Data Summary Dashboard:

Host	Count	Last Update
FIN-WS01	28,277	8/3/26 4:29:15.000 AM
MKT-WS01	2,404	8/3/26 4:43:31.000 AM
ST-DC01	39,526	8/3/26 3:57:38.000 AM

New Search View:

Search query: `host="ST-DC01"`

Time range: Last 24 hours

Events (1,005)

Timeline format

Zoom Out

Format

Show: 20 Per Page

View: List

Selected Fields:

- host
- source
- sourcetype

Interesting Fields:

- Account.Domain
- Account.Name
- collection
- ComputerName
- counter
- EventCode
- EventType
- index
- instance
- Keywords
- linecount
- LogName
- Logon_ID
- Logon_Type
- Message
- object
- OpCode

Event 1:

Time: 8/2/26 7:17:22.985 AM

Event: 08/02/2026 08:17:22.985 -0700

collection="Network Interface"

object="Network Interface"

counter="Bytes Sent/sec"

instance="Intel(R) 82574L Gigabit Network Connection"

host = ST-DC01 source = Perfmon:Network Interface sourcetype = Perfmon:Network Interface

Event 2:

Time: 8/2/26 7:17:22.985 AM

Event: 08/02/2026 08:17:22.985 -0700

collection="Network Interface"

object="Network Interface"

counter="Bytes Received/sec"

instance="Intel(R) 82574L Gigabit Network Connection"

host = ST-DC01 source = Perfmon:Network Interface sourcetype = Perfmon:Network Interface

Event 3:

Time: 8/2/26 7:17:22.985 AM

Event: 08/02/2026 08:17:22.985 -0700

collection="Available Memory"

object="Memory"

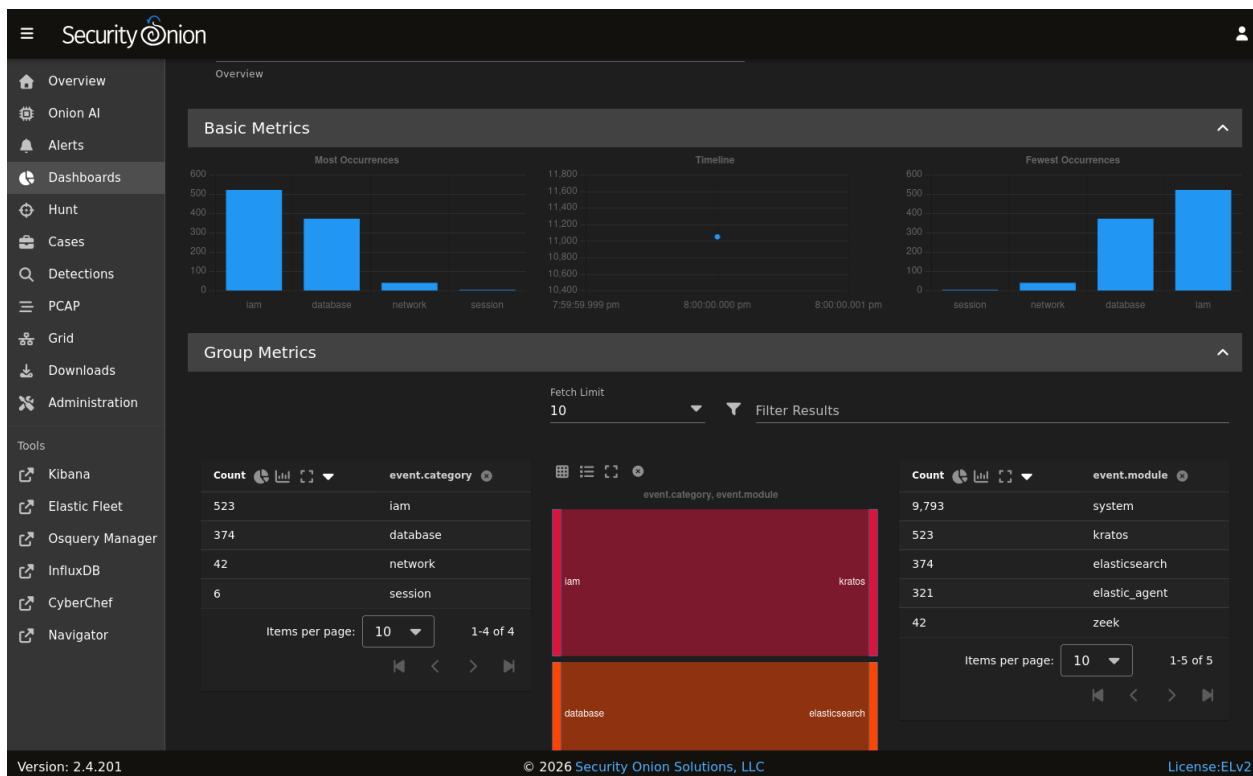
counter="Available Bytes"

instance="0"

host = ST-DC01 source = Perfmon:Available Memory sourcetype = Perfmon:Available Memory

Security Onion

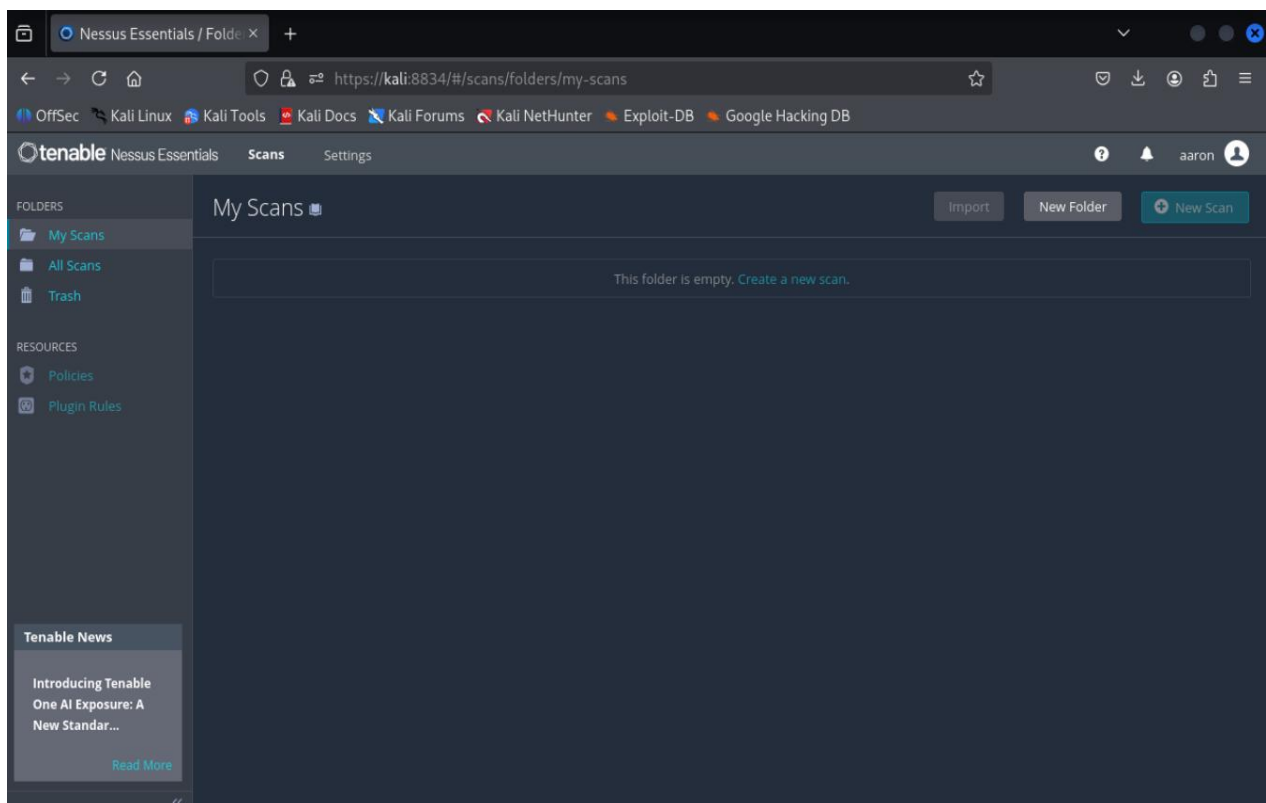
Security Onion serves as the network security monitoring (NSM) platform within the Stellaris Tech simulation environment. It provides centralized visibility into network traffic and security activity by collecting and analyzing network data to identify suspicious or potentially malicious behavior. Security Onion integrates multiple open-source security tools to support network traffic analysis, intrusion detection, packet analysis, and alert generation. Within the environment, Security Onion complements Splunk by providing network-focused monitoring and detection capabilities, while Splunk serves as the primary SIEM for centralized security log analysis and event correlation.



Vulnerability Scanning

Nessus

Nessus is one of the primary vulnerability assessment tools in the Stellaris Tech simulation environment. It is used to identify security vulnerabilities, missing patches, insecure configurations, and other potential weaknesses across systems on the network. Nessus performs authenticated and unauthenticated scans to evaluate the security posture of Windows hosts and other network devices, providing detailed reports that help prioritize and remediate identified risks.



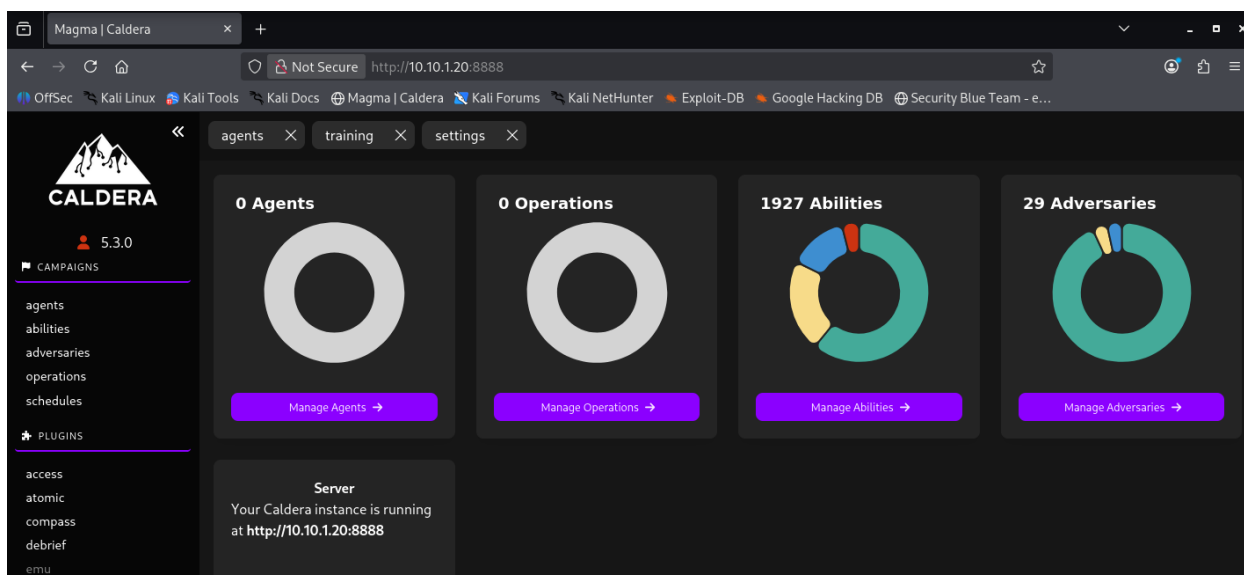
OpenVAS

OpenVAS serves as a complementary vulnerability assessment platform within the Stellaris Tech simulation environment. It identifies security vulnerabilities, missing patches, and configuration weaknesses across systems, providing an additional layer of validation and comparison for Nessus vulnerability assessments. By comparing scan results across both platforms, vulnerabilities can be verified, false positives reduced, and differences in detection capabilities identified. Using two independent vulnerability scanners provides broader assessment coverage and increases confidence in the accuracy of vulnerability findings, supporting a more comprehensive vulnerability management process throughout the environment.

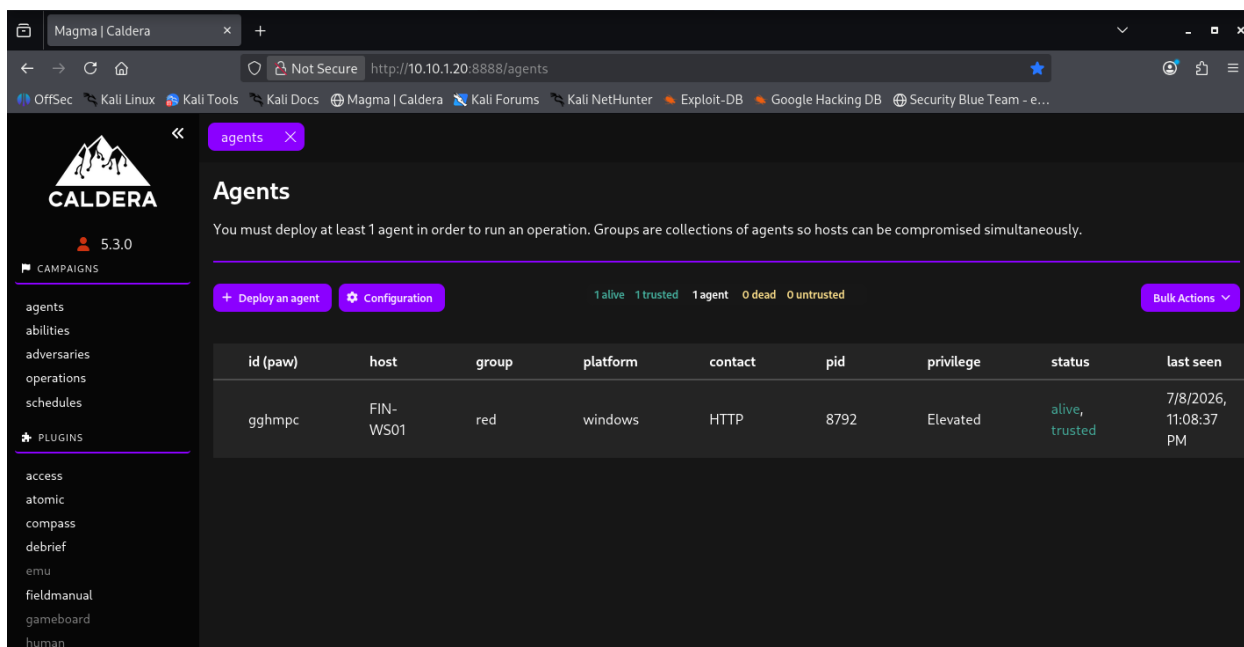
The screenshot displays the OpenVAS web interface. At the top, a black header bar contains the OpenVAS logo, navigation icons, and the user 'admin'. A left sidebar lists menu items: Dashboards, Scans, Assets, Resilience, Security Information, Configuration, Administration, and Help. The main content area features a blue notification banner stating 'Feed is currently syncing.' with a close button. Below this, a 'Dashboards' section is visible, with the 'Overview' tab selected. Two 3D pie charts are shown, both with a title of '(Total: 0)'. The first chart is titled 'Tasks by Severity Class' and the second is 'Tasks by Status'. The version '26.9.0' is noted in the bottom left corner.

MITRE Caldera

MITRE Caldera serves as the adversary emulation platform within the Stellaris Tech simulation environment. It simulates real-world cyberattacks using the MITRE ATT&CK framework, enabling security tools and defensive controls to be tested against realistic attack techniques. Caldera executes automated attack scenarios to generate security events, enabling the validation of detection rules, security monitoring capabilities, and incident response processes across the environment. By emulating adversary behavior, MITRE Caldera provides a controlled method for evaluating the effectiveness of the lab's defensive security tools while supporting threat hunting and security operations training.



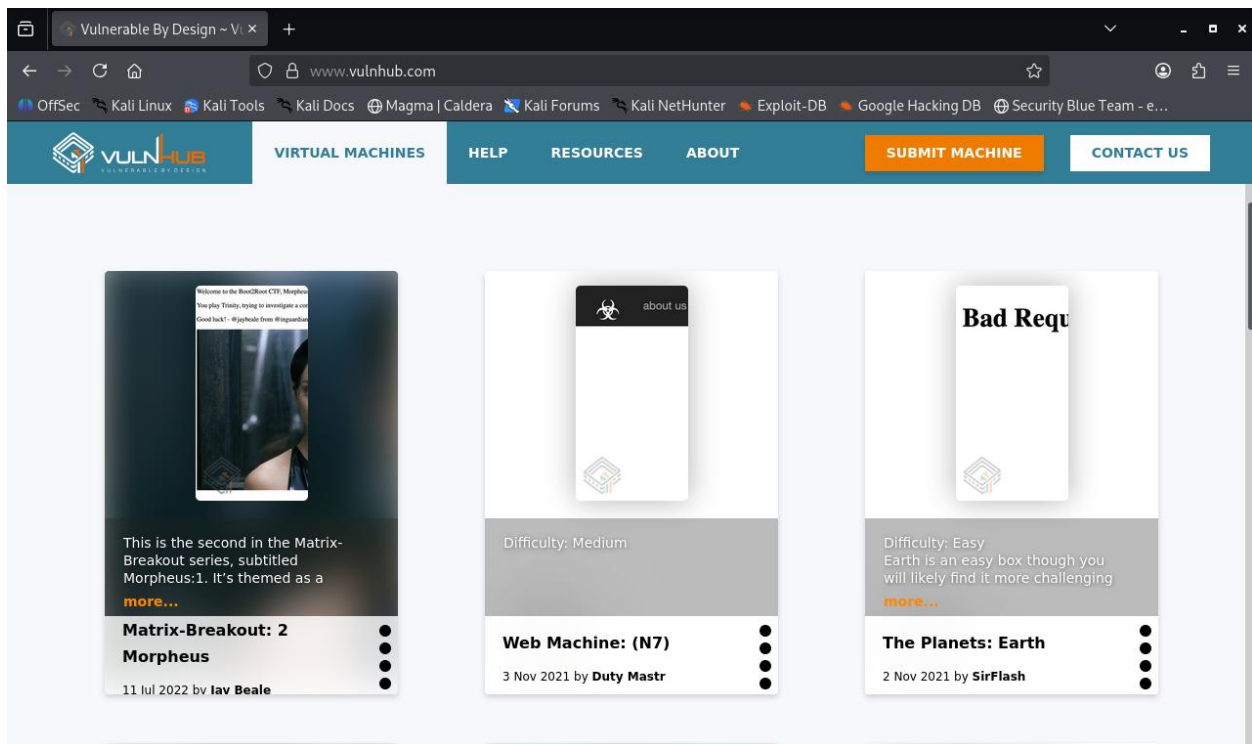
Caldera agent connected to FIN-WS01 is shown below:



Vulnerable Machines

VulnHUB

VulnHub is a website that provides intentionally vulnerable virtual machines that simulate real-world systems with known security weaknesses for cybersecurity training and testing. Within the Stellaris Technologies simulation environment, VulnHub machines are deployed as controlled targets for vulnerability assessments and penetration testing.



Metasploitable 2

Metasploitable 2 is an intentionally vulnerable Linux virtual machine that contains numerous known vulnerabilities and insecure services designed for cybersecurity training and security testing. Within the Stellaris Tech simulation environment, it serves as a target system for penetration testing and threat detection exercises.

[illegible]

Quarantine

Quarantine Overview

The Quarantine subnet is an isolated network segment reserved for systems that require containment during security testing or incident response activities. Its purpose is to separate potentially compromised or high-risk devices from production systems, limiting lateral movement while allowing further investigation and remediation. Although not used for day-to-day operations, the quarantine subnet demonstrates an enterprise security practice by providing a dedicated location for isolating systems during simulated security incidents. The subnet is maintained as a dedicated containment zone for future incident response and security validation exercises.

Conclusion

The Stellaris Tech Simulation Environment provides a realistic enterprise-style cybersecurity lab designed to support hands-on learning across multiple security disciplines, including network administration, Active Directory management, vulnerability assessment, security monitoring, threat detection, and incident response. By combining enterprise infrastructure with industry-standard security tools, the environment serves as a practical platform for developing and validating cybersecurity skills in a controlled setting. As the environment continues to evolve, additional scenarios, detection logic, and defensive capabilities can be incorporated to simulate real-world security operations further and support ongoing professional development.

Skills Learned

Throughout the project, the following technical skills were developed and reinforced:

- Enterprise network architecture and subnet segmentation
- Virtual machine configuration and OS installation
- pfSense firewall configuration and access control
- Active Directory Domain Services (AD DS) administration
- Windows workstation configuration and administration
- Secure Shell (SSH) remote administration and server access
- Sysmon configuration and endpoint telemetry collection
- DNS and DHCP configuration
- Organizational Unit (OU) and role-based access control (RBAC) design
- Windows Server 2022 and Windows workstation administration
- Security documentation and infrastructure design
- Network troubleshooting and configuration analysis
- Principle of Least Privilege
- Security tool deployment and environment management

Future Projects

The Stellaris Tech Simulation Environment will continue to serve as the foundation for hands-on cybersecurity projects that build upon the existing infrastructure. Planned projects include:

- **Splunk Boss of the SOC (BOTS):** Analyze realistic attack datasets to develop Search Processing Language (SPL) skills, investigate Advanced Persistent Threat (APT) activity, and improve threat hunting techniques.
- **Adversary Emulation and Threat Hunting:** Use MITRE Caldera to simulate attacker behavior and generate realistic security events. Analyze the resulting activity using Splunk, Security Onion, and Sysmon telemetry to identify attack techniques, investigate suspicious behavior, document findings, and improve detection capabilities.
- **Detection Engineering:** Develop custom SPL searches, dashboards, and alerts to improve detection capabilities for common attack techniques.
- **Vulnerability Assessment and Remediation:** Perform vulnerability scans using Nessus and OpenVAS, prioritize findings, implement remediation, and validate security improvements.
- **Incident Response:** Investigate simulated security incidents and produce professional incident reports detailing analysis, containment, eradication, and recovery activities.